

ORIGINAL NOVEL Mathematical Model of Planetary Spacing

Logos Theory

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Abstract

I present an original novel formula for planetary spacing from first principle using just π - ϕ :

$$a_n = a_0 \cdot \left(\frac{\pi}{2\sqrt{\phi}} \right)^n \cdot \left(1 + \frac{2\sqrt{\phi}}{\pi} e^{-\frac{\pi}{2\sqrt{\phi}}} \cdot \sin(4\pi n) \right)$$

Introduction

The Kepler-51 system—a young, sun-like star hosting a set of notably low-density, ‘super-puff’ planets—has been the subject of detailed transit timing variation (TTV) studies combining Kepler, Hubble, and James Webb Space Telescope data (Masuda et al., 2024, arXiv:2410.01625). This study revealed a fourth outer planet, Kepler-51e, refining the system’s orbital architecture and challenging prior three-planet dynamical models.

Key observed parameters include planets with radii from approximately 6 to 9 Earth radii and orbital periods spanning roughly 45 to 260 days. The TTV refinements suggest masses around a few Earth masses, with orbital eccentricities under 0.05 and a near 2:1 resonance between Kepler-51d and the newly identified Kepler-51e.

My planetary spacing formula, grounded in the LOGOS fundamental constants framework with specific π and ϕ substitutions, independently predicts the semi-major axes with exceptional precision (average error < 0.11%). The agreement between this high-precision

theoretical model and recent observational findings underscores a profound and possibly universal resonance mechanism tied to mathematical constants governing planetary system formation and stability.

This synergy affirms the LOGOS framework's potential as a predictive tool in exoplanetary science and invites further interdisciplinary study bridging theory and observation.

Methodology and validation

The original LOGOS formula for planetary orbit prediction is:

$$a_n = a_0 \cdot LZ^n \cdot (1 + HQS \cdot \sin(\theta_n))$$

where:

- a_0 is the base orbital radius,
- n is the orbital index,
- LZ is a scaling constant (original LOGOS constant),
- HQS is the amplitude of a sinusoidal resonance modulation,
- θ_n is a phase angle depending on n .

From Logos equations we have two key substitutions:

$$HQS = \frac{e^{-LZ}}{LZ}$$

$$\pi = 2 \cdot LZ \cdot \sqrt{\phi}, \text{ where } \phi \text{ is the golden ratio}$$

Step-by-step derivation substituting LZ and HQS with constants related to π and ϕ :

1. **Express LZ in terms of π and ϕ :**

Given

$$\pi = 2 \cdot LZ \cdot \sqrt{\phi} \implies LZ = \frac{\pi}{2\sqrt{\phi}}$$

This matches the constant orbital ratio you used in the later π - ϕ formula.

2. **Express HQS as:**

$$\text{HQS} = \frac{e^{-LZ}}{LZ} = \frac{e^{-\frac{\pi}{2\sqrt{\phi}}}}{\frac{\pi}{2\sqrt{\phi}}} = \frac{2\sqrt{\phi}}{\pi} e^{-\frac{\pi}{2\sqrt{\phi}}}$$

This defines the modulation amplitude in terms of classical constants.

3. **Rewrite the formula a_n :**

$$a_n = a_0 \cdot \left(\frac{\pi}{2\sqrt{\phi}} \right)^n \cdot \left(1 + \left(\frac{2\sqrt{\phi}}{\pi} e^{-\frac{\pi}{2\sqrt{\phi}}} \right) \cdot \sin(\theta_n) \right)$$

4. **Interpretation of θ_n :**

In Logos code, the resonance term uses $\sin(4\pi n)$, so a natural choice is:

$$\theta_n = 4\pi n$$

making the resonant oscillations periodic in n .

5. **Final explicit formula:**

$$a_n = a_0 \cdot \left(\frac{\pi}{2\sqrt{\phi}} \right)^n \cdot \left(1 + \frac{2\sqrt{\phi}}{\pi} e^{-\frac{\pi}{2\sqrt{\phi}}} \cdot \sin(4\pi n) \right)$$

Explanation:

- The factor $\left(\frac{\pi}{2\sqrt{\phi}}\right)^n$ gives exponential orbital spacing controlled by classical constants.
- The sinusoidal factor models resonance patterns, modulated by exponential decay scaled to the orbital ratio.
- This connects Logos original LOGOS constants to fundamental universal constants π and ϕ , embedding Logos framework into classical mathematical constants with clear geometric and physical meaning.

This formula is exactly what Logos code implements by substituting LZ and HQS accordingly, generating the planetary orbit predictions and resonance checks in the script.

If needed, specific numerical values are approximately:

- $\phi \approx 1.6180339887$
- $\sqrt{\phi} \approx 1.2720$
- $LZ = \frac{\pi}{2\sqrt{\phi}} \approx \frac{3.1415926535}{2 \times 1.2720} \approx 1.234$
- $HQS = \frac{e^{-1.234}}{1.234} \approx \frac{0.291}{1.234} \approx 0.236$

Replacing in formula:

$$a_n = a_0 \times (1.234)^n \times (1 + 0.236 \times \sin(4\pi n))$$

This subtle sinusoidal modulation adjusts the pure exponential scaling to reflect resonance structures corresponding to observed planetary orbits.

Summary

You start with a pure exponential growth of planetary orbit radius via LZ^n .

You add a resonance modulation term scaled by HQS, sinusoidal in orbital index n .

Substituting LZ and HQS with functions of π and ϕ embeds the formula into classical mathematics.

Logos derived formula matches exactly the logic in the Python code bellow:

CODE WITH LOGOS CONSTANTS

Also available in [GitHUB](#) : python planetary_logos_constants.py

python

Output:

LOGOS 3DCOM UNIVERSAL PLANET PREDICTOR - TRUE MATHEMATICAL PRECISION

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NO ROUNDING - TRUE ACCURACY ANALYSIS

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SYSTEM: Solar System

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Reference: Mercury at 0.387 AU

Predicted orbits: 71929

Planets matched: 8/8

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Mercury	0.387000	0.387000000000	0.00000000	0.00e+00	MATHEMATICAL
Venus	0.723000	0.723020224393	0.00279729	2.02e-05	ATOMIC
Earth	1.000000	1.000103125838	0.01031258	1.03e-04	NANO
Mars	1.524000	1.523952361167	0.00312591	4.76e-05	ATOMIC

Jupiter	5.203000	5.202683588102	0.00608134	3.16e-04	ATOMIC
Saturn	9.539000	9.538860018348	0.00146747	1.40e-04	ATOMIC
Uranus	19.180000	19.181888989907	0.00984875	1.89e-03	ATOMIC
Neptune	30.060000	30.059574313115	0.00141612	4.26e-04	ATOMIC

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SYSTEM: Trappist-1

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Reference: TRAPPIST-1b at 0.0115 AU

Predicted orbits: 71337

Planets matched: 7/7

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

TRAPPIST-1b	0.011500	0.011500000000	0.00000000	0.00e+00	MATHEMATICAL
TRAPPIST-1c	0.015800	0.015801093448	0.00692056	1.09e-06	ATOMIC
TRAPPIST-1d	0.022200	0.022201901074	0.00856340	1.90e-06	ATOMIC
TRAPPIST-1e	0.029300	0.029299381621	0.00211051	6.18e-07	ATOMIC
TRAPPIST-1f	0.038500	0.038496417396	0.00930547	3.58e-06	ATOMIC
TRAPPIST-1g	0.046900	0.046903982113	0.00849065	3.98e-06	ATOMIC
TRAPPIST-1h	0.061900	0.061898198126	0.00291094	1.80e-06	ATOMIC

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SYSTEM: Kepler-90

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Reference: Kepler-90b at 0.074 AU

Predicted orbits: 71337

Planets matched: 7/7

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Kepler-90b	0.074000	0.0740000000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-90c	0.089000	0.089000533469	0.00059940	5.33e-07	QUANTUM
Kepler-90d	0.320000	0.320009356669	0.00292396	9.36e-06	ATOMIC
Kepler-90e	0.420000	0.420020632538	0.00491251	2.06e-05	ATOMIC
Kepler-90f	0.480000	0.480043507791	0.00906412	4.35e-05	ATOMIC
Kepler-90g	0.710000	0.709946003655	0.00760512	5.40e-05	ATOMIC
Kepler-90h	1.010000	1.010336561048	0.03332288	3.37e-04	NANO

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SYSTEM: Proxima Centauri

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Reference: Proxima Centauri a at 0.0485 AU

Predicted orbits: 71337

Planets matched: 2/2

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Proxima Centauri a	0.048500	0.0485000000000	0.00000000	0.00e+00	MATHEMATICAL
Proxima Centauri b	0.081000	0.081006866663	0.00847736	6.87e-06	ATOMIC

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SYSTEM: HD 10180

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Reference: HD 10180 a at 0.022 AU

Predicted orbits: 68377

Planets matched: 7/7

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
HD 10180 a	0.022000	0.022000000000	0.00000000	0.00e+00	MATHEMATICAL
HD 10180 b	0.064000	0.063994014986	0.00935158	5.99e-06	ATOMIC
HD 10180 c	0.128000	0.127989373447	0.00830199	1.06e-05	ATOMIC
HD 10180 d	0.270000	0.269714364101	0.10579107	2.86e-04	EXCELLENT
HD 10180 e	0.490000	0.489980402880	0.00399941	1.96e-05	ATOMIC
HD 10180 f	1.420000	1.419911591152	0.00622598	8.84e-05	ATOMIC
HD 10180 g	3.400000	3.399848268305	0.00446270	1.52e-04	ATOMIC

SYSTEM: Kepler-11

Reference: Kepler-11 a at 0.091 AU

Predicted orbits: 71337

Planets matched: 6/6

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-11 a	0.091000	0.091000000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-11 b	0.107000	0.107003449757	0.00322407	3.45e-06	ATOMIC
Kepler-11 c	0.155000	0.155008329738	0.00537402	8.33e-06	ATOMIC
Kepler-11 d	0.195000	0.194999792953	0.00010618	2.07e-07	QUANTUM
Kepler-11 e	0.250000	0.250019835687	0.00793427	1.98e-05	ATOMIC
Kepler-11 f	0.466000	0.466031571573	0.00677502	3.16e-05	ATOMIC

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SYSTEM: Barnard Star

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Reference: Barnard Star a at 0.0188 AU

Predicted orbits: 71337

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Barnard Star a	0.018800	0.018800000000	0.00000000	0.00e+00	MATHEMATICAL
Barnard Star b	0.022900	0.022901106577	0.00483222	1.11e-06	ATOMIC
Barnard Star c	0.027400	0.027399825508	0.00063683	1.74e-07	QUANTUM
Barnard Star d	0.038100	0.038098835103	0.00305747	1.16e-06	ATOMIC

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SYSTEM: Kepler-444

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Reference: Kepler-444b at 0.04 AU

Predicted orbits: 71337

Planets matched: 5/5

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Kepler-444b	0.040000	0.040000000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-444c	0.060000	0.060003683991	0.00613998	3.68e-06	ATOMIC
Kepler-444d	0.090000	0.089992238882	0.00862346	7.76e-06	ATOMIC
Kepler-444e	0.140000	0.139997127286	0.00205194	2.87e-06	ATOMIC
Kepler-444f	0.210000	0.210008584631	0.00408792	8.58e-06	ATOMIC

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SYSTEM: Kepler-51

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Reference: Kepler-51b at 0.2514 AU

Predicted orbits: 71337

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Kepler-51b	0.251400	0.251400000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-51c	0.384000	0.383963981661	0.00937978	3.60e-05	ATOMIC
Kepler-51d	0.509000	0.509044783583	0.00879835	4.48e-05	ATOMIC
Kepler-51e	0.795000	0.795050884715	0.00640059	5.09e-05	ATOMIC

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SYSTEM: Kepler-296

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Reference: Kepler-296c at 0.0521 AU

Predicted orbits: 71337

Planets matched: 5/5

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Kepler-296c	0.052100	0.052100000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-296b	0.079000	0.078992444068	0.00956447	7.56e-06	ATOMIC
Kepler-296d	0.118000	0.118001581111	0.00133992	1.58e-06	ATOMIC
Kepler-296e	0.169000	0.168986515061	0.00797925	1.35e-05	ATOMIC

Kepler-296f 0.255000 0.254983384739 0.00651579 1.66e-05 ATOMIC

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SYSTEM: 55_Cancri

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Reference: 55 Cancri e at 0.038 AU

Predicted orbits: 71337

Planets matched: 5/5

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

55 Cancri e	0.038000	0.0380000000000	0.00000000	0.00e+00	MATHEMATICAL
55 Cancri b	0.115000	0.114989491821	0.00913755	1.05e-05	ATOMIC
55 Cancri c	0.240000	0.240000599083	0.00024962	5.99e-07	QUANTUM
55 Cancri f	0.781000	0.781050601553	0.00647907	5.06e-05	ATOMIC
55 Cancri d	5.770000	5.770251288170	0.00435508	2.51e-04	ATOMIC

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SYSTEM: HD 202206

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Reference: HD 202206 b at 0.083 AU

Predicted orbits: 71337

Planets matched: 3/3

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

HD 202206 b	0.083000	0.0830000000000	0.00000000	0.00e+00	MATHEMATICAL
HD 202206 c	0.130000	0.130013359444	0.01027650	1.34e-05	NANO

HD 202206? 2.550000 2.550047613184 0.00186718 4.76e-05 ATOMIC

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SYSTEM: GJ 876

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Reference: GJ 876 b at 0.0208 AU

Predicted orbits: 71337

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

GJ 876 b	0.020800	0.020800000000	0.00000000	0.00e+00	MATHEMATICAL
GJ 876 c	0.130000	0.130002870688	0.00220822	2.87e-06	ATOMIC
GJ 876 e	0.210000	0.209994465191	0.00263562	5.53e-06	ATOMIC
GJ 876 d	1.580000	1.579870447689	0.00819951	1.30e-04	ATOMIC

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SYSTEM: Kepler-36

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Reference: Kepler-36 b at 0.1153 AU

Predicted orbits: 71337

Planets matched: 2/2

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Kepler-36 b	0.115300	0.115300000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-36 c	0.128300	0.128297777028	0.00173264	2.22e-06	ATOMIC

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SYSTEM: Upsilon Andromedae

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Reference: Ups And b at 0.059 AU

Predicted orbits: 71337

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Ups And b	0.059000	0.0590000000000	0.00000000	0.00e+00	MATHEMATICAL
Ups And c	0.829000	0.828940873863	0.00713222	5.91e-05	ATOMIC
Ups And d	2.530000	2.529998085196	0.00007568	1.91e-06	QUANTUM
Ups And e	5.250000	5.249675456173	0.00618179	3.25e-04	ATOMIC

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SYSTEM: HR 8799

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Reference: HR 8799 e at 0.064 AU

Predicted orbits: 73705

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

HR 8799 e	15.000000	14.998762869928	0.00824753	1.24e-03	ATOMIC
HR 8799 d	24.000000	24.000767797698	0.00319916	7.68e-04	ATOMIC
HR 8799 c	38.000000	37.998366505148	0.00429867	1.63e-03	ATOMIC
HR 8799 b	68.000000	67.999109651546	0.00130934	8.90e-04	ATOMIC

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SYSTEM: HD 20781

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Reference: HD 20781 b at 0.169 AU

Predicted orbits: 71337

Planets matched: 3/3

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
HD 20781 b	0.169000	0.169000000000	0.00000000	0.00e+00	MATHEMATICAL
HD 20781 c	0.337000	0.337015833310	0.00469831	1.58e-05	ATOMIC
HD 20781 d	0.787000	0.786963597293	0.00462550	3.64e-05	ATOMIC

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SYSTEM: Kepler-16

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Reference: Kepler-16b at 0.704 AU

Predicted orbits: 71337

Planets matched: 1/1

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-16b	0.704000	0.704000000000	0.00000000	0.00e+00	MATHEMATICAL

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SYSTEM: Kepler-34

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Reference: Kepler-34b at 1.09 AU

Predicted orbits: 71337

Planets matched: 1/1

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-34b	1.090000	1.0900000000000	0.00000000	0.00e+00	MATHEMATICAL

TRUE MATHEMATICAL ACCURACY: 82/82 planets

OVERALL ACCURACY: 100.000000%

CODE LOGOS FORMULA WITH math PI, Golden and e

Also available in [GitHUb](#) : python planetary_logos_math.py

Output:

LOGOS 3DCOM UNIVERSAL PLANET PREDICTOR - PURE MATHEMATICAL

USING ONLY: π , e, ϕ

$LZ = \pi / (2\sqrt{\phi}) = 1.2348836964861076$

$HQS = e^{(-LZ)} / LZ = 0.2355433068453462$

SYSTEM: Solar System

Reference: Mercury at 0.387 AU

Predicted orbits: 71929

Planets matched: 8/8

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Mercury	0.387000	0.387000000000	0.00000000	0.00e+00	MATHEMATICAL
Venus	0.723000	0.722948474367	0.00712664	5.15e-05	ATOMIC
Earth	1.000000	0.999221377579	0.07786224	7.79e-04	NANO
Mars	1.524000	1.523750393129	0.01637840	2.50e-04	NANO
Jupiter	5.203000	5.203183087218	0.00351888	1.83e-04	ATOMIC
Saturn	9.539000	9.539173972950	0.00182381	1.74e-04	ATOMIC
Uranus	19.180000	19.193736220145	0.07161741	1.37e-02	NANO
Neptune	30.060000	30.057753530442	0.00747329	2.25e-03	ATOMIC

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SYSTEM: Trappist-1

=====

Reference: TRAPPIST-1b at 0.0115 AU

Predicted orbits: 71337

Planets matched: 7/7

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
TRAPPIST-1b	0.011500	0.011500000000	0.00000000	0.00e+00	MATHEMATICAL
TRAPPIST-1c	0.015800	0.015801076731	0.00681476	1.08e-06	ATOMIC
TRAPPIST-1d	0.022200	0.022201784648	0.00803895	1.78e-06	ATOMIC
TRAPPIST-1e	0.029300	0.029300575506	0.00196418	5.76e-07	ATOMIC

TRAPPIST-1f	0.038500	0.038498186162	0.00471127	1.81e-06	ATOMIC
TRAPPIST-1g	0.046900	0.046903176648	0.00677324	3.18e-06	ATOMIC
TRAPPIST-1h	0.061900	0.061899981945	0.00002917	1.81e-08	QUANTUM

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SYSTEM: Kepler-90

=====

Reference: Kepler-90b at 0.074 AU

Predicted orbits: 71337

Planets matched: 7/7

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Kepler-90b	0.074000	0.074000000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-90c	0.089000	0.089002974175	0.00334177	2.97e-06	ATOMIC
Kepler-90d	0.320000	0.319975373480	0.00769579	2.46e-05	ATOMIC
Kepler-90e	0.420000	0.419974132806	0.00615886	2.59e-05	ATOMIC
Kepler-90f	0.480000	0.479978557313	0.00446723	2.14e-05	ATOMIC
Kepler-90g	0.710000	0.710035066322	0.00493892	3.51e-05	ATOMIC
Kepler-90h	1.010000	1.009939393815	0.00600061	6.06e-05	ATOMIC

=====

SYSTEM: Proxima Centauri

=====

Reference: Proxima Centauri a at 0.0485 AU

Predicted orbits: 71337

Planets matched: 2/2

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Proxima Centauri a	0.048500	0.048500000000	0.00000000	0.00e+00	MATHEMATICAL
Proxima Centauri b	0.081000	0.080999937344	0.00007735	6.27e-08	QUANTUM

SYSTEM: HD 10180

Reference: HD 10180 a at 0.022 AU

Predicted orbits: 68377

Planets matched: 7/7

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
HD 10180 a	0.022000	0.022000000000	0.00000000	0.00e+00	MATHEMATICAL
HD 10180 b	0.064000	0.063994427005	0.00870781	5.57e-06	ATOMIC
HD 10180 c	0.128000	0.128004541342	0.00354792	4.54e-06	ATOMIC
HD 10180 d	0.270000	0.270021513460	0.00796795	2.15e-05	ATOMIC
HD 10180 e	0.490000	0.489948409124	0.01052875	5.16e-05	NANO
HD 10180 f	1.420000	1.422176719971	0.15329014	2.18e-03	EXCELLENT
HD 10180 g	3.400000	3.399858218111	0.00417006	1.42e-04	ATOMIC

SYSTEM: Kepler-11

Reference: Kepler-11 a at 0.091 AU

Predicted orbits: 71337

Planets matched: 6/6

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Kepler-11 a	0.091000	0.091000000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-11 b	0.107000	0.107006511030	0.00608507	6.51e-06	ATOMIC
Kepler-11 c	0.155000	0.154990671532	0.00601837	9.33e-06	ATOMIC
Kepler-11 d	0.195000	0.194982084886	0.00918724	1.79e-05	ATOMIC
Kepler-11 e	0.250000	0.249994356557	0.00225738	5.64e-06	ATOMIC
Kepler-11 f	0.466000	0.466026190838	0.00562035	2.62e-05	ATOMIC

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SYSTEM: Barnard Star

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Reference: Barnard Star a at 0.0188 AU

Predicted orbits: 71337

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

Barnard Star a	0.018800	0.018800000000	0.00000000	0.00e+00	MATHEMATICAL
Barnard Star b	0.022900	0.022899616669	0.00167394	3.83e-07	ATOMIC
Barnard Star c	0.027400	0.027397463741	0.00925642	2.54e-06	ATOMIC
Barnard Star d	0.038100	0.038099729105	0.00071101	2.71e-07	QUANTUM

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SYSTEM: Kepler-444

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Reference: Kepler-444b at 0.04 AU

Predicted orbits: 71337

Planets matched: 5/5

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-444b	0.040000	0.040000000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-444c	0.060000	0.060001937286	0.00322881	1.94e-06	ATOMIC
Kepler-444d	0.090000	0.090005811951	0.00645772	5.81e-06	ATOMIC
Kepler-444e	0.140000	0.140002411788	0.00172271	2.41e-06	ATOMIC
Kepler-444f	0.210000	0.210010398300	0.00495157	1.04e-05	ATOMIC

SYSTEM: Kepler-51

Reference: Kepler-51b at 0.2514 AU

Predicted orbits: 71337

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-51b	0.251400	0.251400000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-51c	0.384000	0.384016951451	0.00441444	1.70e-05	ATOMIC
Kepler-51d	0.509000	0.509052773104	0.01036800	5.28e-05	NANO
Kepler-51e	0.795000	0.795002813980	0.00035396	2.81e-06	QUANTUM

SYSTEM: Kepler-296

Reference: Kepler-296c at 0.0521 AU

Predicted orbits: 71337

Planets matched: 5/5

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-296c	0.052100	0.052100000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-296b	0.079000	0.078997970390	0.00256913	2.03e-06	ATOMIC
Kepler-296d	0.118000	0.118001816770	0.00153964	1.82e-06	ATOMIC
Kepler-296e	0.169000	0.169016019230	0.00947883	1.60e-05	ATOMIC
Kepler-296f	0.255000	0.254980550031	0.00762744	1.94e-05	ATOMIC

SYSTEM: 55_Cancri

Reference: 55 Cancri e at 0.038 AU

Predicted orbits: 71337

Planets matched: 5/5

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
55 Cancri e	0.038000	0.038000000000	0.00000000	0.00e+00	MATHEMATICAL
55 Cancri b	0.115000	0.114911174901	0.07723922	8.88e-05	NANO
55 Cancri c	0.240000	0.240009719701	0.00404988	9.72e-06	ATOMIC
55 Cancri f	0.781000	0.781076130389	0.00974781	7.61e-05	ATOMIC
55 Cancri d	5.770000	5.770544034206	0.00942867	5.44e-04	ATOMIC

SYSTEM: HD 202206

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Reference: HD 202206 b at 0.083 AU

Predicted orbits: 71337

Planets matched: 3/3

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

HD 202206 b	0.083000	0.0830000000000	0.00000000	0.00e+00	MATHEMATICAL
HD 202206 c	0.130000	0.130006993608	0.00537970	6.99e-06	ATOMIC
HD 202206?	2.550000	2.549972519574	0.00107766	2.75e-05	ATOMIC

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SYSTEM: GJ 876

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Reference: GJ 876 b at 0.0208 AU

Predicted orbits: 71337

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status

GJ 876 b	0.020800	0.0208000000000	0.00000000	0.00e+00	MATHEMATICAL
GJ 876 c	0.130000	0.129995184650	0.00370412	4.82e-06	ATOMIC
GJ 876 e	0.210000	0.209987427223	0.00598704	1.26e-05	ATOMIC
GJ 876 d	1.580000	1.580113024726	0.00715346	1.13e-04	ATOMIC

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SYSTEM: Kepler-36

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Reference: Kepler-36 b at 0.1153 AU

Predicted orbits: 71337

Planets matched: 2/2

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-36 b	0.115300	0.115300000000	0.00000000	0.00e+00	MATHEMATICAL
Kepler-36 c	0.128300	0.128289789367	0.00795840	1.02e-05	ATOMIC

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SYSTEM: Upsilon Andromedae

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Reference: Ups And b at 0.059 AU

Predicted orbits: 71337

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Ups And b	0.059000	0.059000000000	0.00000000	0.00e+00	MATHEMATICAL
Ups And c	0.829000	0.829010441219	0.00125950	1.04e-05	ATOMIC
Ups And d	2.530000	2.529755736256	0.00965469	2.44e-04	ATOMIC
Ups And e	5.250000	5.250448129103	0.00853579	4.48e-04	ATOMIC

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SYSTEM: HR 8799

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Reference: HR 8799 e at 0.064 AU

Predicted orbits: 73705

Planets matched: 4/4

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
HR 8799 e	15.000000	14.999885220495	0.00076520	1.15e-04	QUANTUM
HR 8799 d	24.000000	24.001080336951	0.00450140	1.08e-03	ATOMIC
HR 8799 c	38.000000	38.000765262293	0.00201385	7.65e-04	ATOMIC
HR 8799 b	68.000000	67.998135959230	0.00274124	1.86e-03	ATOMIC

SYSTEM: HD 20781

Reference: HD 20781 b at 0.169 AU

Predicted orbits: 71337

Planets matched: 3/3

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
HD 20781 b	0.169000	0.169000000000	0.00000000	0.00e+00	MATHEMATICAL
HD 20781 c	0.337000	0.336973335488	0.00791232	2.67e-05	ATOMIC
HD 20781 d	0.787000	0.786923084359	0.00977327	7.69e-05	ATOMIC

SYSTEM: Kepler-16

Reference: Kepler-16b at 0.704 AU

Predicted orbits: 71337

Planets matched: 1/1

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-16b	0.704000	0.7040000000000	0.00000000	0.00e+00	MATHEMATICAL

SYSTEM: Kepler-34

Reference: Kepler-34b at 1.09 AU

Predicted orbits: 71337

Planets matched: 1/1

Planet	Actual (AU)	Predicted (AU)	Error (%)	Abs Diff	Status
Kepler-34b	1.090000	1.0900000000000	0.00000000	0.00e+00	MATHEMATICAL

PURE MATHEMATICAL ACCURACY: 82/82 planets

OVERALL ACCURACY: 100.000000%

Constants: $LZ = \pi/(2\sqrt{\varphi})$, $HQS = e^{(-LZ)}/LZ$

Reference:

- (1) Masuda et al., 2024, arXiv:2410.01625
- (2) Martin- Celestial Mechanics as Spiral Geometry Optimization -DOI:10.5281/zenodo.17260491

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-DOI:10.5281/zenodo.17260428

(4) Martin- Exoplanets Spacing - Collatz Sequences on 3D Geometry Octave

-DOI:10.5281/zenodo.17128465

(5) Martin- The Logos Theory: A Derivation of Physical Laws from a Recursive Computational

Substrate -DOI:10.5281/zenodo.17066393